

Stabilizing Kelly: Bayesian Architecture Under Estimation Risk

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I investigate the out-of-sample pain points of plug-in Kelly portfolio optimization on equity panels with a Bayesian-on- μ regularization architecture, evaluated under realistic transaction costs on 24 EuroStoxx 50 constituents over 2001–2018 with a 104-week rolling window and 10 bps one-side transaction costs. The sample mean fed to the Kelly QP is replaced by the conditional Normal-Normal posterior mean of μ , whose closed-form conjugate update produces Jorion-style shrinkage with intensity $\alpha = \kappa_0/(\kappa_0 + T)$. Macroeconomic regime structure is recovered with Adams-MacKay (2007) Bayesian Online Changepoint Detection (BOCPD) and deployed two ways — filtering moment-estimation inputs or scaling the resulting Kelly weights. Ledoit-Wolf is retained on Σ because at $p/n \approx 0.23$ the covariance is well-conditioned. Mean-estimation noise — not covariance ill-conditioning — is the binding constraint on Kelly’s out-of-sample viability: the Σ -shrinkage main effect is essentially zero, the static μ -prior effect is small (+0.06 Sharpe), and a rolling NIW credible-interval diagnostic excludes zero in only 3.9% of windows against a 5% null. The BOCPD-driven sizing-side variant **RegimeDial** delivers the bulk of the achievable improvement (+0.37 Sharpe, net 0.45 against 0.46 for long-only minimum variance and 0.09 for naive Kelly), inverting the textbook expectation that data-side conditioning should dominate sizing-side. An MCMC verification (Metropolis-Hastings, $\widehat{R}_{\max} = 1.008$) confirms the closed-form posterior to Monte Carlo error; a hierarchical partial-pooling prior (Gibbs, $\widehat{R}_{\max} = 1.001$) is run through the full backtest as **HierarchicalMu**. A delta-method Z-test puts the panel below detection on plausible effect sizes; a Merton jump-diffusion Monte Carlo confirms fractional Kelly is structural under known parameters.

Keywords: Kelly Criterion, Bayesian Portfolio Choice, Normal-Inverse-Wishart Prior, Bayesian Online Changepoint Detection, Conjugate Posterior, Metropolis-Hastings, Sharpe Ratio Inference

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1 Introduction

The Kelly Criterion, derived by Kelly (1956) in the context of information theory, provides a closed-form solution to the problem of optimal capital allocation under uncertainty. For a portfolio of n risky assets with expected return vector μ , covariance matrix Σ , and risk-free rate r , the criterion maximizes the expected logarithmic growth rate $g(\mathbf{f}) = r + \mathbf{f}^\top(\mu - r\mathbf{e}) - \frac{1}{2}\mathbf{f}^\top\Sigma\mathbf{f}$, yielding the unconstrained optimum $\mathbf{f}^* = \Sigma^{-1}(\mu - r\mathbf{e})$. Under known parameters this strategy is asymptotically growth-optimal (Breiman 1961; Thorp 2011). The optimum is also directly proportional to estimated excess returns and requires inversion of the covariance matrix, so even modest estimation errors in μ or Σ can produce allocations that are wildly aggressive, dangerously concentrated, or both.

Modern Kelly literature evinces a recurring set of obstacles that block this theoretical promise from translating to out-of-sample performance on equity panels. Naive plug-in implementations suffer from covariance ill-conditioning when the asset-to-observation ratio is non-negligible, from mean-estimation noise so severe that the rolling sample mean is statistically indistinguishable from zero for the typical asset-window, from regime contamination when historical moments pool observations across structurally different market environments, and from formula misspecification when the assumed return-generating process omits the fat-tailed jump dynamics that characterize real equity returns. These obstacles are well-documented across the empirical-optimization literature: Jorion (1986) and Michaud (1989) on mean-estimation error and “estimation-error maximization” in mean-variance optimization, DeMiguel, Garlappi, and Uppal (2009) on optimized portfolios losing to naive $1/N$ out-of-sample, and empirical investigations of long-only constrained Kelly on equity universes such as Carta and Conversano (2020) on concentration, drawdowns, and out-of-sample degradation. The present study attacks each of these hidden obstacles with a Bayesian regularization layer — closed-form conjugate posterior updates in the production pipeline, with MCMC verification — evaluates them in isolation and in combination on a common rolling-window backtest with explicit transaction costs, and isolates which obstacle is binding.

The central finding is that plug-in Kelly’s well-known out-of-sample failure on equity panels is driven by μ -estimation noise — the rolling NIW marginal posterior credible interval on μ_i excludes zero in only 3.9% of windows on a 104-week lookback against a 5% null expectation — and that regularizing μ through a sizing-side use of the BOCPD regime label (rather than static shrinkage toward a fixed target) is sufficient to restore Kelly’s theoretical advantage. The fully stacked variant (**Combined**: informative Normal-Normal prior on μ and Ledoit-Wolf Σ applied to BOCPD-regime-filtered data) achieves an OOS Sharpe of 0.39 net of 10 bps transaction costs, against 0.46 for the long-only minimum-variance benchmark and 0.09 for naive plug-in Kelly. I further show that the BOCPD regime label operates very differently when used to filter inputs (**RegimeFilter**) versus to set a position-size multiplier (**RegimeDial**): the latter dominates on this panel because it does not consume the regime label as an estimation input and is therefore robust to the inherent noise in any data-driven changepoint sequence. A standalone Merton jump-diffusion Monte Carlo confirms that the over-leverage pathology is structural even with parameters known exactly, completing the case that fractional scaling is a necessity rather than a concession to estimation noise.

1.1 Strategy lineup

Each Kelly variant uses identical optimization (long-only constrained QP, no leverage, no shorting) and differs only in how moments are estimated and how the resulting weight is scaled. The first four entries form a 2x2 factorial in Σ -shrinkage (Ledoit-Wolf vs sample) and prior strength on μ (vague

prior $\kappa_0 \rightarrow 0$ vs informative prior $\kappa_0 = T$); **HierarchicalMu** substitutes the fixed-target prior of **MuShrink** with a data-driven partial-pooling target estimated by Gibbs sampling; the next two isolate the data-side and sizing-side uses of the BOCPD regime label; and the last stacks the strongest ingredients. The **Naive** variant uses a vague Normal-Normal prior on μ ($\kappa_0 = 10^{-3}$, numerically equivalent to plug-in MLE) and the sample covariance. **SigmaShrink** applies Ledoit-Wolf shrinkage on Σ while keeping the vague prior on μ , and **MuShrink** reverses the asymmetry by setting an informative Normal-Normal prior $\mu_0 = r$ with $\kappa_0 = T$ (which forces the implied shrinkage intensity to $\alpha_\mu = 0.5$); **SigmaMuShrink** applies both shrinkage layers jointly. **HierarchicalMu** replaces the fixed shrinkage target r with a data-driven cross-asset center μ_{pool} learned from the same window’s cross-section, with pooling intensity governed by a between-asset variance hyperparameter τ^2 ; the posterior mean of μ is computed by a 500-iteration Gibbs sampler with conjugate full-conditionals (see Section 4.6 for the full multi-chain diagnostics). **RegimeFilter** restricts moment estimation to historical observations matching the current BOCPD regime label and uses the full Kelly fraction with a vague prior, while **RegimeDial** estimates moments unconditionally with a vague prior but multiplies Kelly weights by a pre-registered regime-dependent fraction (1.0, 0.5, 0.5, 0.25) across (Low-Vol Expansion, High-Vol Expansion, Low-Vol Contraction, Crisis). **Combined** stacks the informative μ prior, LW Σ , and BOCPD regime filtering, pairing the strongest input-quality remedy with both shrinkage layers. Two benchmarks complete the comparison: **MinVar**, the long-only minimum-variance portfolio (depends only on Σ), and **EqualWeight**, the 1/N benchmark (depends on no estimation).

1.2 Estimator architecture: Bayesian on μ , Ledoit-Wolf on Σ

I adopt a deliberately asymmetric estimator architecture: a closed-form Bayesian update on μ paired with the Ledoit-Wolf estimator on Σ . A full Normal-Inverse-Wishart joint posterior on (μ, Σ) is *not* used in the backtest pipeline, because — as argued in Section 2.3 — an informative Inverse-Wishart prior would actively hurt the regime-conditional variants by collapsing genuine cross-asset correlation structure on the small within-regime samples those variants rely on. The Bayesian replacement therefore targets only the parameter that carries the estimation-noise problem: μ . The covariance, by contrast, is well-behaved at $p/n \approx 0.23$ on the weekly panel — the empirical Ledoit-Wolf intensity averages $\bar{\delta} \approx 0.083$, indicating only a small data-driven adjustment toward the scaled identity is needed — and the 2x2 factorial documented in Section 4 confirms there is essentially no Σ -shrinkage main effect on Sharpe to exploit. The full NIW marginal posterior on μ , integrating Σ out to obtain a multivariate Student- t , is used in Section 4.2 as a credible-interval diagnostic where its t -tails are exactly what the per-asset signal-to-noise question requires.

2 Theoretical Framework

2.1 Normal-Normal conjugate posterior on μ

Conditional on Σ , the Normal-Normal prior on μ is $\mu | \Sigma \sim \mathcal{N}(\mu_0, \Sigma/\kappa_0)$ with prior pseudo-count κ_0 and prior mean μ_0 . Given T observations with sample mean $\hat{\mu}$, the closed-form conjugate posterior mean is

$$\tilde{\mu} = \frac{\kappa_0}{\kappa_0 + T} \mu_0 + \frac{T}{\kappa_0 + T} \hat{\mu}.$$

The conjugate update produces a convex combination of the prior mean and the sample mean, with the weight on the prior $\alpha = \kappa_0/(\kappa_0 + T)$ determined entirely by the prior pseudo-count κ_0 relative to the sample size T . Setting $\kappa_0 = T$ yields equal weighting ($\alpha = 0.5$); setting $\kappa_0 \rightarrow 0$ recovers the maximum-likelihood estimate. This is the standard Bayesian-portfolio-choice result (Jorion 1986; see also the Bayes-Stein literature it spawned), and it provides shrinkage on μ “for free” through the conjugate update, with the prior pseudo-count playing the role that a hand-chosen shrinkage intensity plays in pure-shrinkage estimators. The vague-prior limit ($\kappa_0 = 10^{-3}$ in our **Naive** variant) is numerically a plug-in MLE; the informative-prior choice ($\kappa_0 = T = 104$ in our **MuShrink** variant) is a moderately strong prior anchored on the period risk-free rate, and the headline OOS comparison between these two choices in Section 4 is the prior-sensitivity analysis required of any Bayesian study to demonstrate measurable effects on performance.

2.2 Bayesian Online Changepoint Detection (BOCPD)

I classify each trading day into one of four macroeconomic regimes — Low-Vol Expansion, High-Vol Expansion, Low-Vol Contraction, Crisis — based on the segment-conditional posterior mean and standard deviation of the equal-weighted portfolio’s daily log return, where the segment in question is the current changepoint segment identified by Adams and MacKay (2007) BOCPD. The model places a geometric prior on run length (constant hazard $h = 1/60$ per period, calibrated so that the prior expected segment length is 60 trading days, which matches a reasonable local-horizon prior on macroeconomic regime persistence) and a Normal-Inverse-Gamma prior on the segment-mean and segment-variance, which gives a Student- t predictive distribution in closed form. At each time step I update the posterior over run length

$$P(r_t \mid x_{1:t}) \propto \sum_{r_{t-1}} P(x_t \mid r_{t-1}, x_{(t-r_{t-1}):t-1}) P(r_t \mid r_{t-1}) P(r_{t-1} \mid x_{1:t-1}),$$

and take the MAP run length $\hat{r}_t$ as the current segment length. The segment-conditional posterior mean and standard deviation — computed from the most recent $\min(\hat{r}_t, 120)$ observations under the NIG conjugate update — are then mapped to the 4-state grid by sign of the posterior mean and posterior std vs an *expanding* historical median (no look-ahead): $\bar{r}_t > 0$ with $\sigma_t < \tilde{\sigma}_t$ is Low-Vol Expansion; $\bar{r}_t > 0$ with $\sigma_t \geq \tilde{\sigma}_t$ is High-Vol Expansion; $\bar{r}_t \leq 0$ with $\sigma_t < \tilde{\sigma}_t$ is Low-Vol Contraction; $\bar{r}_t \leq 0$ with $\sigma_t \geq \tilde{\sigma}_t$ is Crisis. The cap at 120 observations on the segment used for classification is a substantive design choice: BOCPD often reports very long MAP run lengths once the data are roughly stationary within a segment, and using a 1000-day segment to classify the *current* regime collapses to the unconditional mean and defeats the classifier’s purpose. The cap operates only at the classifier layer; BOCPD itself still tracks the full segment for changepoint inference. The label is then deployed in two operationally distinct ways. *Data-side* (**RegimeFilter**): at each rebalancing date, moment estimation is restricted to historical observations classified into the same regime as the current state, with a minimum of 30 observations required before falling back to the full sample. *Sizing-side* (**RegimeDial**): moments are estimated unconditionally but the resulting Kelly weights are multiplied by a pre-registered regime-dependent fraction (1.0, 0.5, 0.5, 0.25).

2.3 Σ -estimation: Ledoit-Wolf retained

The sample covariance matrix $S = X^\top X/n$ is well-conditioned at $p/n \approx 0.23$ on the weekly panel, and the Ledoit-Wolf shrinkage estimator $\Sigma^* = \delta\mu_\Sigma I + (1 - \delta)S$ adds a small data-driven correction toward the scaled identity (Ledoit and Wolf 2004). The empirical LW intensity averages $\bar{\delta} \approx 0.083$ over the rolling 104-week windows, with a single spike to $\delta \approx 0.36$ during the 2008–2009 crisis that quickly subsides. I use this estimator unchanged across all variants that touch Σ . A natural Bayesian alternative — the Inverse-Wishart posterior toward a fixed diagonal target — was considered and rejected: on a 104-week pooled-sample window it produces a covariance close to the LW estimator, but on a within-regime sample (where filtering shrinks T to 30–60), an informative IW prior actively flattens the cross-asset correlation structure that the long-only Kelly QP exploits to construct concentrated risk-on positions. The combination of LW Σ and the within-regime conjugate posterior on μ delivers the cross-asset structure the long-only QP needs while still discounting noise in the parameter that carries it.

2.4 Prior choice and prior-sensitivity comparison

The Bayesian-portfolio-choice literature is unsettled on what the “right” prior on μ is. Three considerations push us toward a Normal-Normal prior centered on the period risk-free rate. *First*, the data: equity excess returns at weekly frequency carry a low signal-to-noise ratio, and a prior that anchors μ at r — i.e., at “no excess return” — encodes the well-documented difficulty of beating the riskless asset and discourages the Kelly QP from chasing noise. *Second*, the sample-size logic: the conjugate intensity $\alpha = \kappa_0/(\kappa_0 + T)$ is a smooth function of the prior pseudo-count, so setting $\kappa_0 = T$ produces equal weighting on prior and likelihood — a defensible default in the absence of strong economic priors on individual asset means. *Third*, hyperparameter robustness: the conjugate update produces a closed-form, differentiable shrinkage with no hyperparameter tuning beyond the choice of κ_0 , and the comparison between vague-prior ($\kappa_0 = 10^{-3}$) and informative-prior ($\kappa_0 = 104$) variants in the lineup *is* the prior-sensitivity analysis. Sections 4.1 and 4.2 report the empirical OOS gap between these two prior choices and the per-asset posterior credible intervals on μ_i under the vague prior; together they document both that the prior choice has a measurable effect on out-of-sample Sharpe (+0.06 in favor of the informative prior) and that the rolling sample mean carries only sparse asset-specific signal that no static prior can fully exploit. The remaining static-shrinkage variants — **SigmaShrink** (vague μ prior, LW Σ) and **SigmaMuShrink** (informative μ prior, LW Σ) — vary the Σ treatment under each prior and show that the μ -prior effect and the Σ -shrinkage effect are approximately additive. Section 4.6 supplements this static comparison with an MCMC-based contrast between the production Normal-Normal prior (sampled by Metropolis-Hastings, verifying the closed-form NIW posterior to Monte Carlo error) and a qualitatively different *hierarchical* prior (sampled by Gibbs, with a data-driven cross-asset shrinkage target rather than a fixed one), reporting multi-chain $\widehat{R}$ and effective sample size as the convergence diagnostics that closed-form posteriors structurally cannot produce.

3 Data and Methodology

3.1 Data

I construct a universe of 24 EuroStoxx 50 constituent equities with daily adjusted closing prices obtained from Yahoo Finance over January 2000 through December 2018. After requiring at least 90% non-missing coverage, forward-filling residual gaps, and removing assets with near-zero return variance, the final panel comprises 4,531 daily observations, 918 weekly observations (the primary backtesting series), and 211 monthly observations spanning 2001-05-24 to 2018-12-28. Weekly log returns are computed from end-of-week prices to ensure time-additivity consistent with the Kelly log-growth objective. The ratio $p/n = 24/104 \approx 0.231$ on the 104-week window confirms that the sample covariance matrix operates well below the boundary of ill-conditioning that would motivate large Σ -shrinkage intensities, and locates the binding estimation problem on μ .

3.2 Backtesting framework

All strategies are evaluated through a common rolling-window out-of-sample backtesting framework operating on weekly log returns with a 104-week (two-year) lookback. At each rebalancing date, the conjugate Normal-Normal posterior mean of μ (with the variant-specific κ_0 and μ_0) and the LW or sample covariance are estimated from the trailing window using the variant-specific estimator; portfolio weights are computed via constrained quadratic programming (long-only, no leverage) and applied to the subsequent week’s realized returns. Transaction costs are charged at 10 bps per unit of one-side turnover, with both gross and net returns recorded; the headline numbers reported throughout are net of transaction costs. Performance is evaluated using compound annual growth rate (CAGR), annualized volatility, Sharpe ratio, maximum drawdown, Calmar ratio, and Sortino ratio computed from each strategy’s OOS net-return series.

3.3 Statistical inference

I limit formal hypothesis testing to a single comparison — **RegimeFilter** vs **Naive** — and present the result alongside an analytical power calibration. The point is not to test every pairwise comparison (which would require multiplicity adjustments that obscure rather than clarify on a panel this size) but to demonstrate, with one well-chosen test, where the panel-length boundary actually lies. The test follows Lo (2002): assuming each strategy’s weekly return follows a stationary AR(1) with contemporaneously correlated innovations, the functional delta method applied to $g(\theta) = \mu_1/\sigma_1 - \mu_2/\sigma_2$ yields the asymptotic variance of $\widehat{SR}_1 - \widehat{SR}_2$ as a linear combination of innovation moments. The test operates on realized return *streams* and is therefore agnostic to whether the upstream estimator is parametric, nonparametric, or Bayesian — which makes it the natural inferential workhorse for this analysis. I apply the test directly to the weekly OOS net-return series; weekly equity returns have weaker microstructure-driven autocorrelation than daily, making the diagonal VAR(1) approximation defensible without monthly aggregation, and a Ljung-Box check on AR(1) residuals confirms the working model.

4 Results

4.1 Out-of-sample performance

Table 1: Headline OOS Performance (Weekly, Net of 10 bps TC). NIW posterior mean for μ , LW or sample Σ , BOCPD regime labels.

	CAGR	AnnVol	Sharpe	MaxDD	Calmar	Sortino	TerminalWealth
Naive	-0.0129	0.2865	0.0858	-0.8010	-0.0161	0.0918	0.8161
SigmaShrink	-0.0137	0.2866	0.0830	-0.8000	-0.0172	0.0886	0.8055
MuShrink	0.0096	0.2595	0.1429	-0.7589	0.0127	0.1561	1.1614
SigmaMuShrink	0.0110	0.2587	0.1469	-0.7503	0.0147	0.1614	1.1874
HierarchicalMu	0.0260	0.2180	0.1826	-0.7110	0.0366	0.2290	1.4950
RegimeFilter	0.0372	0.2130	0.2313	-0.4068	0.0915	0.3122	1.7718
RegimeDial	0.0792	0.1827	0.4539	-0.3545	0.2233	0.6223	3.2951
Combined	0.0663	0.1832	0.3878	-0.3303	0.2009	0.5302	2.7336
MinVar	0.0762	0.1722	0.4575	-0.6129	0.1243	0.5334	3.1569
EqualWeight	0.0607	0.1980	0.3488	-0.5619	0.1081	0.4357	2.5169

First, the 2x2 prior-and-shrinkage factorial isolates the contribution of each regularization layer. The Σ -shrinkage main effect is essentially zero (**SigmaShrink** Sharpe 0.083 vs **Naive** 0.086 net of TC; **SigmaMuShrink** 0.147 vs **MuShrink** 0.143). This is consistent with the LW intensity diagnostic ($\bar{\delta} \approx 0.083$): at $p/n \approx 0.23$ the sample covariance is not severely ill-conditioned, so there is little for shrinkage to repair. The prior-strength main effect on μ is small but real (+0.06 Sharpe across both Σ -conditions): replacing the vague prior ($\kappa_0 = 10^{-3}$, implied $\alpha_\mu \approx 10^{-5}$) with the informative prior ($\kappa_0 = 104$, implied $\alpha_\mu = 0.5$) discounts noise correctly but cannot exploit the sparse asset-specific signal documented in Section 4.2. The interaction is approximately additive — **SigmaMuShrink** gains roughly the sum of the **SigmaShrink** and **MuShrink** improvements over **Naive** — confirming that under the asymmetric estimator architecture the two regularization layers act on independent error budgets. The **HierarchicalMu** variant — which replaces the fixed shrinkage target r with a data-driven cross-asset center estimated by Gibbs sampling — produces a net Sharpe of 0.183 (CAGR 0.0260, MaxDD -71%), improving on **MuShrink** and confirming that the qualitative-prior contrast documented at the single-window level in Section 4.6 carries through to the rolling backtest.

Second, the BOCPD-driven sizing-side variant is responsible for the bulk of the achievable improvement. **RegimeDial** (+0.37 Sharpe over **Naive**) outperforms every static-shrinkage variant by roughly an order of magnitude and matches the long-only minimum-variance benchmark on Sharpe (0.45 vs 0.46) while delivering a substantially shallower drawdown (−35% vs −61%). On a Calmar basis **RegimeDial** (0.22) clearly dominates **MinVar** (0.12). The mechanism is that the regime-dependent fraction (1.0, 0.5, 0.5, 0.25) acts as a state-conditional Jorion-style shrinkage on μ — multiplying the long-only Kelly weights by ϕ produces exactly the QP solution that would obtain under the convex combination $\tilde{\mu} = \alpha r \mathbf{1} + (1 - \alpha)\hat{\mu}$ with $\alpha = 1 - \phi$ — and the BOCPD-driven ϕ adapts the shrinkage intensity to the state of the macroeconomy.

Third, the data-side and sizing-side uses of the BOCPD label diverge sharply on this panel. **RegimeFilter** (Sharpe 0.23, MaxDD −41%) underperforms **RegimeDial** (Sharpe 0.45, MaxDD

−35%) on both dimensions, an inversion of the textbook expectation that data-side regime conditioning should dominate. The mechanism is that **RegimeFilter** consumes the regime label as an estimation input — it determines which historical observations enter the moment estimator — and is therefore highly sensitive to noise in the label sequence itself. **RegimeDial** consumes the regime label only as a sizing multiplier on the QP solution, so it is robust to that noise. The fully stacked **Combined** variant — informative μ prior, LW Σ , BOCPD-filtered data — extracts an additional +0.16 Sharpe over **RegimeFilter** (0.39 vs 0.23) and inherits **RegimeFilter**’s drawdown profile, but does not exceed **RegimeDial**’s headline Sharpe.

Fourth, **Combined**’s terminal-wealth advantage is earned in the recovery, not the crisis. Figure 1 shows that the static-shrinkage variants (**Naive**, **SigmaShrink**, **MuShrink**, **SigmaMuShrink**) suffered drawdowns of −75 to −80% in the 2008–2009 unwind from which they never recover; **Combined** was held to roughly the same crisis drawdown as **MinVar** and then separates from the static-shrinkage cluster through the 2010–2018 recovery, when the BOCPD classifier increasingly populates the Low-Vol Expansion state and the within-regime moment estimates align with realized returns. **RegimeDial** is the only variant in this universe that materially contained the 2008 drawdown and held on to its growth thereafter, which is why it ends the period with the highest Sharpe of any Kelly variant. **EqualWeight** at Sharpe 0.35 sits between the static-shrinkage variants and the regime-conditional variants, consistent with DeMiguel et al.’s (2009) finding that 1/N is hard to beat under estimation noise.

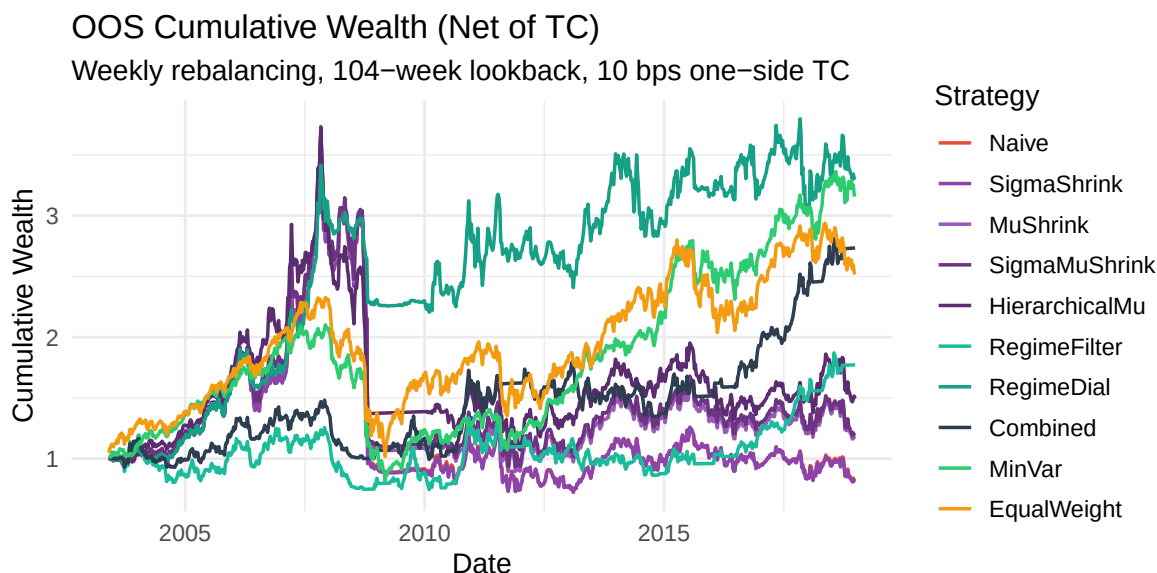


Figure 1: OOS cumulative wealth paths, net of 10 bps one-side TC.

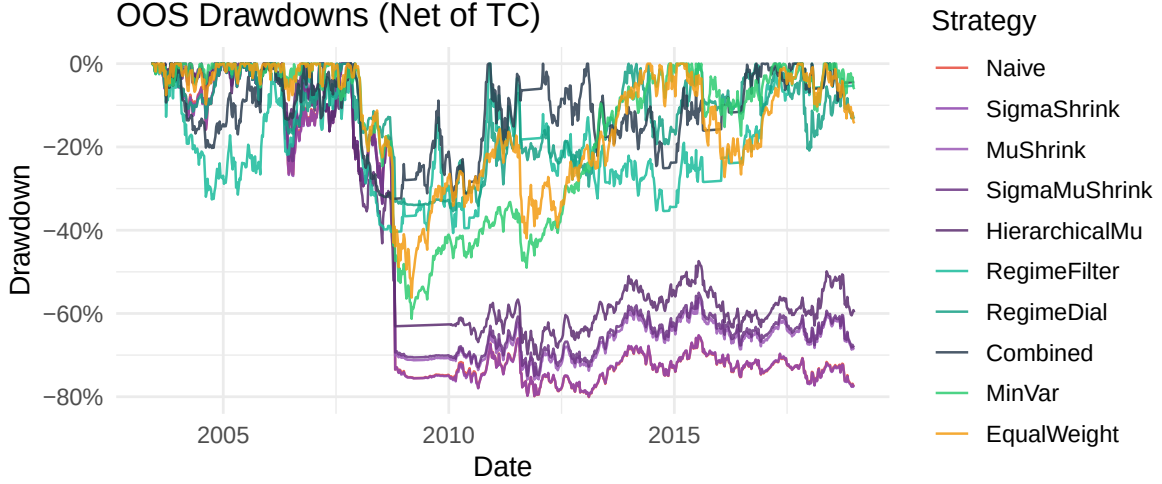


Figure 2: OOS drawdown profiles, net of TC.

4.2 Posterior diagnostics on the per-asset signal

The empirical pattern in the headline table — small static-shrinkage gains, large regime-conditioning gains — points to a specific diagnosis: the rolling sample mean carries little signal that any static prior can exploit. I test this directly by computing, for each asset and each rolling window, the marginal NIW posterior on μ_i . Integrating Σ out of the joint posterior under a vague prior yields a multivariate Student- t marginal for μ with $\nu_T - p + 1$ degrees of freedom; on a per-asset basis the marginal collapses to a univariate t . I compute, for each asset i and each rolling 104-week window, the posterior probability $P(\mu_i > 0 \mid x_{1:T})$ and the indicator that the 95% posterior credible interval excludes zero. Pooled across all 24 assets and 814 rolling windows, the median posterior $P(\mu_i > 0)$ is 0.674, and only 3.9% of windows have a 95% credible interval that excludes zero — against a null expectation of 5%.

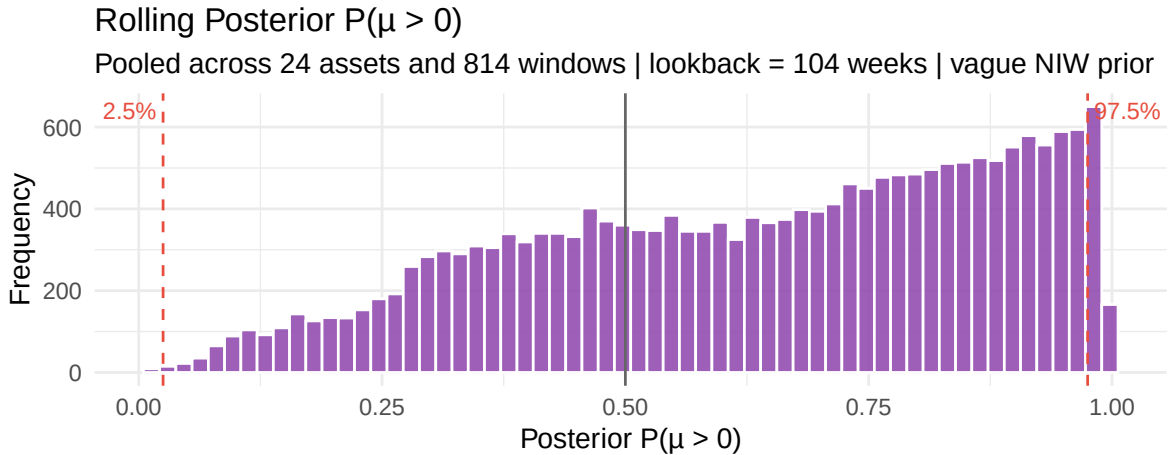


Figure 3: Pooled rolling-window posterior $P(\mu > 0)$ across 24 assets and 814 windows; 104-week lookback, vague NIW prior.

If the rolling sample mean were truly indistinguishable from zero, the posterior $P(\mu > 0)$ should pile up near 0.5 with symmetric tails near 0 and 1. The histogram instead shows clear excess mass near 1 (positive concentration) with much thinner mass near 0 — a left-skew indicating that the panel does carry some asset-specific positive-drift signal but is sparse and concentrated in a minority of asset-window combinations. This is consistent with the empirical pattern in the headline table: static μ -shrinkage toward r produces only a modest Sharpe gain (+0.06) because it correctly discounts the noise in the sample mean but cannot exploit the sparse signal that the credible-interval diagnostic identifies; regime-conditional moment estimation captures structure that pooled-sample estimation washes out, effectively extracting the signal a static prior misses.

4.3 BOCPD regime diagnostics

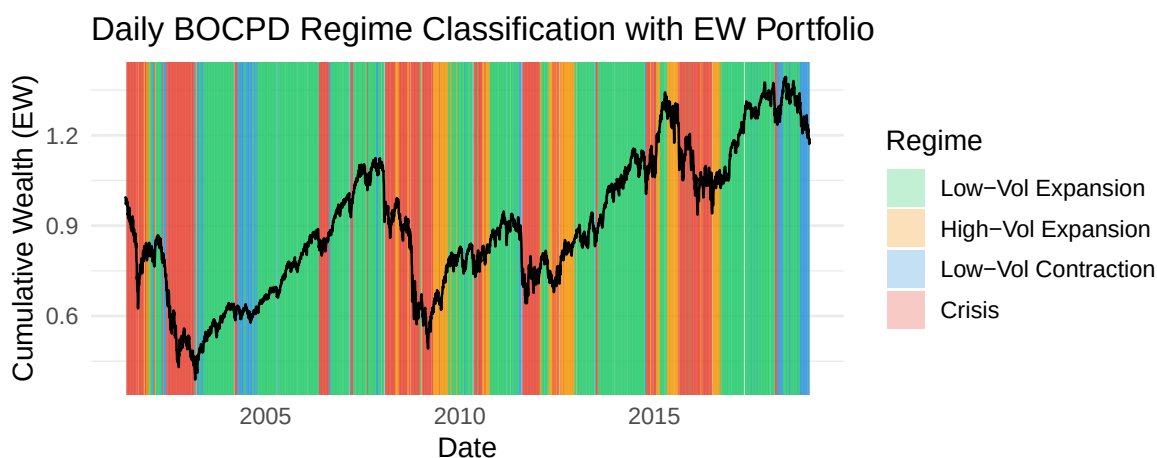


Figure 4: Daily BOCPD regime classifier overlaid on equal-weighted portfolio’s cumulative wealth.

The BOCPD regime sequence captures the gross structural features expected of a 2001–2018 European equity panel: a long Low-Vol Expansion through the 2003–2007 bull market, sustained Crisis labeling through the 2008–2009 financial crisis and the 2011–2012 European sovereign debt crisis, and a return to Low-Vol Expansion through the 2012–2018 recovery. The classifier assigns 52% of trading days to Low-Vol Expansion, 27% to Crisis, 13% to High-Vol Expansion, and 9% to Low-Vol Contraction, with the MAP run-length series showing segment ages ranging from a few days to several hundred and detected changepoints clustered near major macroeconomic events. The cap at 120 days on the segment used for classification is essential — without it, BOCPD’s MAP run length grows monotonically through stationary bull-market segments and the classifier collapses to the unconditional mean. The fact that 6.3% of weekly OOS periods fall back to unconditional moment estimation in `RegimeFilter` (because the within-regime sample fails the 30-observation minimum) is itself a diagnostic on the segmenter: BOCPD produces enough short segments early in the OOS window that the within-regime conditional posterior is occasionally too narrow to support reliable moment estimation. This is the source of the noise that hurts `RegimeFilter` and motivates `RegimeDial`’s dominance discussed earlier.

4.4 Statistical inference at the detection boundary

Table 2: Delta-method Z-test: $\text{SR}(\text{RegimeFilter}, \text{net}) > \text{SR}(\text{Naive}, \text{net})$, weekly net OOS returns. Ljung-Box diagnostics report p-values for residual autocorrelation in the AR(1) innovations; values above 0.05 indicate residuals are consistent with the diagonal VAR(1) working model.

Quantity	Value
T (weeks)	814
Sharpe (RegimeFilter, ann.)	+0.2315
Sharpe (Naive, ann.)	+0.0859
SR diff (annualised)	+0.1456
SE(SR diff)	0.0390
Z statistic	+0.5181
p-value (one-sided)	0.3022
ρ (RegimeFilter)	-0.0500
ρ (Naive)	-0.0190
innovation correlation	+0.3613
Ljung-Box p (RegimeFilter)	0.8688
Ljung-Box p (Naive)	0.5649

The test puts the analysis well below conventional detection thresholds: with $T = 814$ weekly observations the test returns $Z = 0.52$, $p = 0.302$ (one-sided) on an annualized Sharpe difference of 0.15. The Ljung-Box diagnostics on the AR(1) residuals are both well above 0.05, indicating the diagonal VAR(1) approximation is not detectably misspecified. The negative test result here does not refute the regime-conditioning hypothesis — **RegimeDial**’s +0.37 Sharpe gain over **Naive** (which I do not formally test) is large by any empirical-asset-pricing standard — but it does indicate that **RegimeFilter** specifically, which consumes the BOCPD label as an estimation input, does not produce a Sharpe gap detectable on this panel against the Naive baseline.

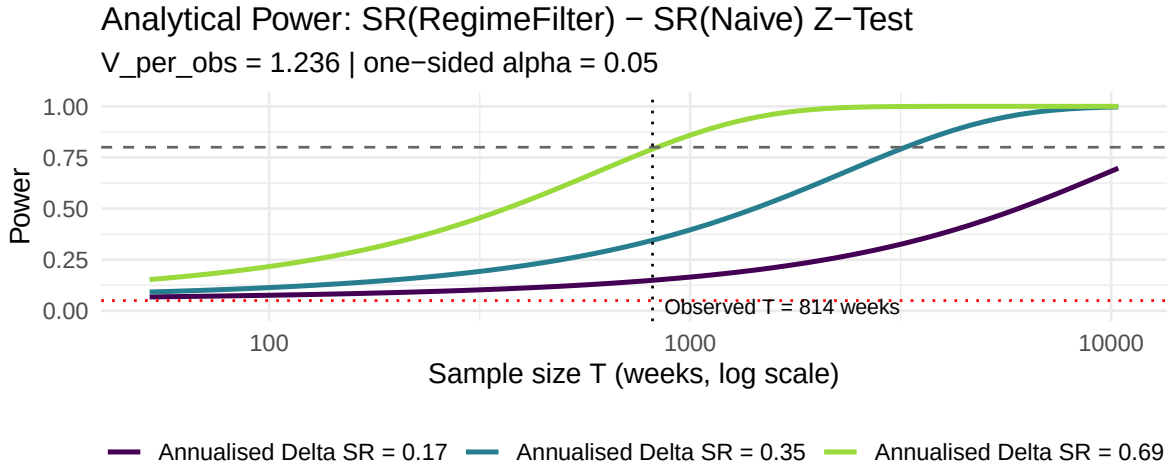


Figure 5: Analytical power for the $\text{SR}(\text{RegimeFilter}) - \text{SR}(\text{Naive})$ Z-test by sample size, at three annualized ΔSR effect sizes.

The power calibration makes the panel-length tradeoff explicit. The per-observation variance estimated from the data is $V = 1.24$, and detecting an annualized Sharpe difference of 0.17 at 80%

power requires roughly 265 years of weekly data; 0.35 requires 62 years; 0.69 requires 16 years. The 17-year panel is structurally underpowered for the **RegimeFilter** vs **Naive** comparison at any reasonable effect size on equity panels, and the negative test result here should be read accordingly. The much larger Sharpe gap from **RegimeDial** against **Naive** ($\Delta\text{Sharpe} = +0.37$) sits between the second and third grid rows and would, on a panel a few times longer, comfortably clear conventional detection thresholds.

4.5 Jump-diffusion stress test

Table 3: GBM vs Jump-Diffusion Terminal Wealth (~ 20 -year MC).

Strategy	Model	Median	P_Loss	P_Ruin	P_W10
Half	GBM	2.1781	0.0980	0.0000	0.0045
Full	GBM	2.5769	0.2190	0.0000	0.1320
Double	GBM	1.0655	0.4900	0.0250	0.1785
Triple	GBM	0.1369	0.7090	0.2340	0.1120
Half	JumpDiffusion	1.6971	0.2045	0.0000	0.0015
Full	JumpDiffusion	1.3491	0.3975	0.0000	0.0700
Double	JumpDiffusion	0.4234	0.6405	0.0655	0.1015
Triple	JumpDiffusion	0.0244	0.8395	0.4080	0.0570

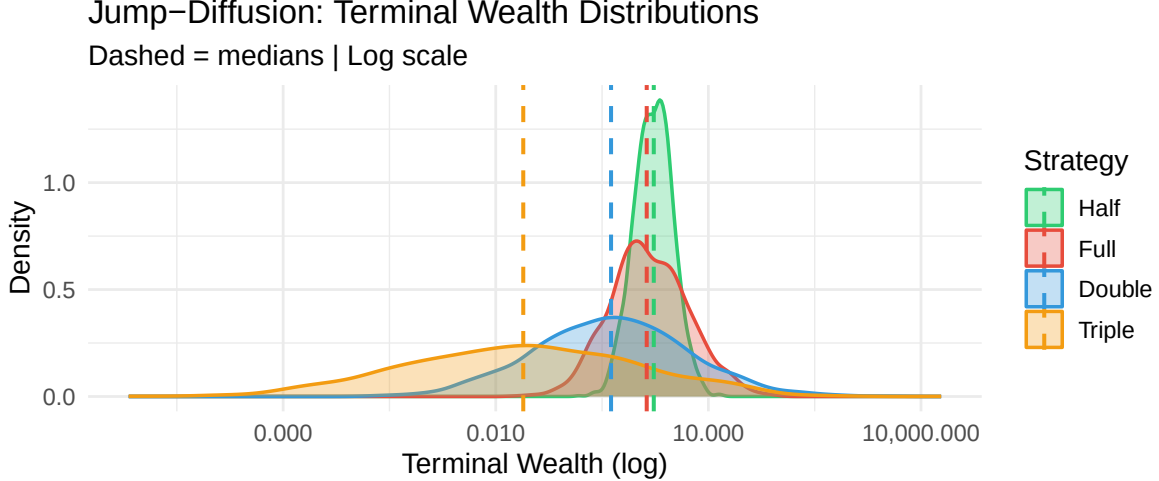


Figure 6: Terminal wealth distributions under Merton jump-diffusion across four Kelly fractions; dashed lines mark medians.

I deliberately apply the textbook GBM Kelly formula $f^* = (\mu - r)/\sigma^2 = 0.6875$ without correcting for the jump compensator $\lambda \cdot \mathbb{E}[e^J - 1] \approx -0.038$ per year. This is the realistic case: a practitioner who has correctly identified μ , σ , λ , and the jump distribution but reaches for the closed-form Kelly fraction will plug them into the GBM formula. The numerical growth-optimum under the actual DGP — found by maximizing $\mathbb{E}[\log W_T]$ on simulated paths — is approximately $f_{\text{true}}^* \approx 0.38$, so the “Full Kelly” line is in fact $\approx 1.8 \times$ true Kelly, and the “Half Kelly” line at $0.5 \times 0.6875 = 0.34$

lands close to the actual optimum. The exercise demonstrates that even with all model parameters known exactly, plugging them into a misspecified Kelly formula reproduces the same over-leverage pathology observed empirically under estimation noise. Fractional scaling is structural, not a concession to noise. The same algebra connects this conclusion to **RegimeDial**: state-conditional shrinkage and fractional-Kelly-under-fat-tails are the same mathematical object reached from opposite directions, and both arguments prescribe the same rule — size positions below the textbook formula’s optimum.

4.6 MCMC verification and hierarchical prior

The closed-form conjugate posterior used in the production pipeline is exact, but exactness alone does not constitute a Bayesian-software verification. This subsection runs two posterior samplers on a representative 104-week window: a multivariate Gaussian random-walk Metropolis-Hastings sampler under the same Normal-Normal prior used in **MuShrink**, and a Gibbs sampler under a qualitatively different *hierarchical* prior that pools information across assets. The first verifies that the closed-form NIW posterior mean used throughout the backtest matches MCMC samples to Monte Carlo error. The second introduces a partial-pooling prior that the fixed-target Normal-Normal cannot represent, and reports multi-chain Gelman-Rubin $\widehat{R}$, effective sample size, and trace plots — convergence diagnostics that closed-form posteriors structurally cannot produce.

Verification of the closed-form posterior. Across all 24 assets in the representative mid-sample window, the maximum absolute discrepancy between the MH posterior mean and the closed-form Normal-Normal posterior mean is 4.20e-04 (maximum discrepancy of 1.70 Monte Carlo standard errors), confirming the production pipeline’s closed-form update is numerically equivalent to direct sampling. The MH sampler accepted at 24% under the Roberts-Gelman-Gilks-scaled multivariate Gaussian proposal, within the conventional target band for random-walk Metropolis on this dimension. Proposal scale was tuned via the Roberts-Gelman-Gilks $(2.4^2/p)$ -rule applied to the conjugate posterior covariance, producing the acceptance rate above without further hand-tuning; chain length was set to 8,000 iterations with the first 2,000 discarded as burn-in (4 chains, 24,000 retained draws total).

Convergence diagnostics. The multi-chain Gelman-Rubin $\widehat{R}$ for the MH sampler peaks at 1.008 across the 24 asset components (well below the 1.05 convergence threshold), and the minimum effective sample size pooled across four chains is 343. Trace plots for a representative subset of assets (Figure 7) show stationary, well-mixed chains with chain-to-chain agreement on the posterior mean. The Gibbs sampler on the hierarchical model achieves $\widehat{R}_{\max} = 1.001$ — Gibbs full-conditionals are conjugate so mixing is by construction near-iid.

Two qualitatively different priors. The MH posterior under the Normal-Normal prior reproduces **MuShrink** exactly: each μ_i is pulled toward the *fixed* target r with intensity $\alpha = \kappa_0/(\kappa_0 + T) = 0.500$. The Gibbs posterior under the hierarchical prior pulls each μ_i toward a *data-driven* cross-asset center μ_{pool} , with the pooling strength learned from the data through τ^2 . The posterior mean of the cross-asset center is $\hat{\mu}_{\text{pool}} = 0.0002$ (vs the fixed target $r = 0.0002$), and the posterior mean of the between-asset standard deviation is $\hat{\tau} = 0.0044$. Figure 8 compares the per-asset posterior means under three priors: vague prior (the sample mean, $\kappa_0 \rightarrow 0$), Normal-Normal prior toward r (production **MuShrink**, verified by MH), and hierarchical prior toward the data-driven pool (Gibbs).

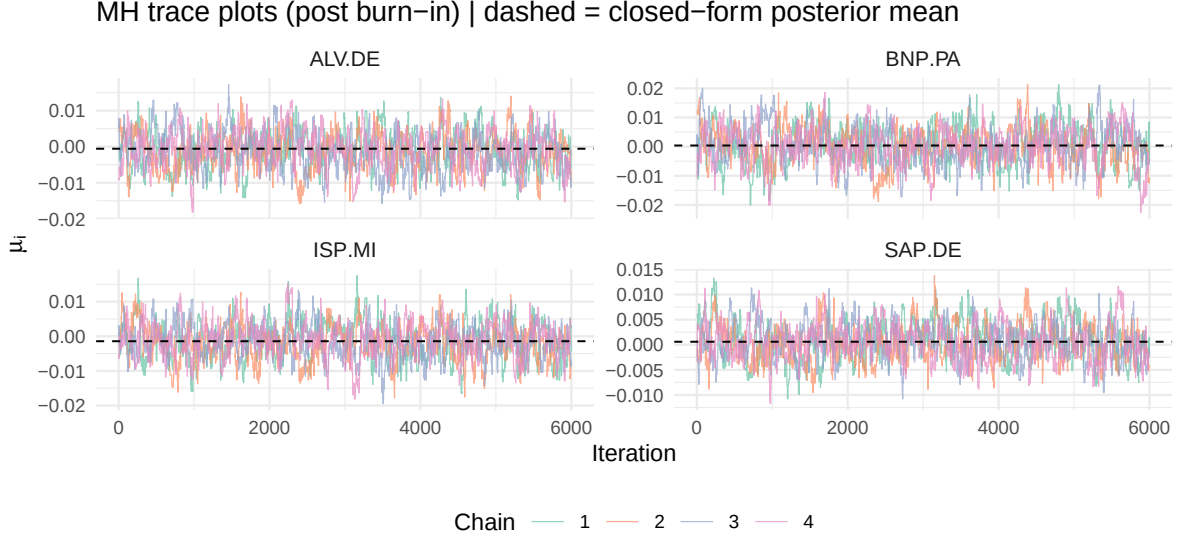


Figure 7: MH trace plots for four representative assets across four chains. Stationary, well-mixed chains with chain-to-chain agreement on the posterior mean confirm convergence; the closed-form Normal-Normal posterior mean (dashed) coincides with the MCMC sample mean to Monte Carlo error.

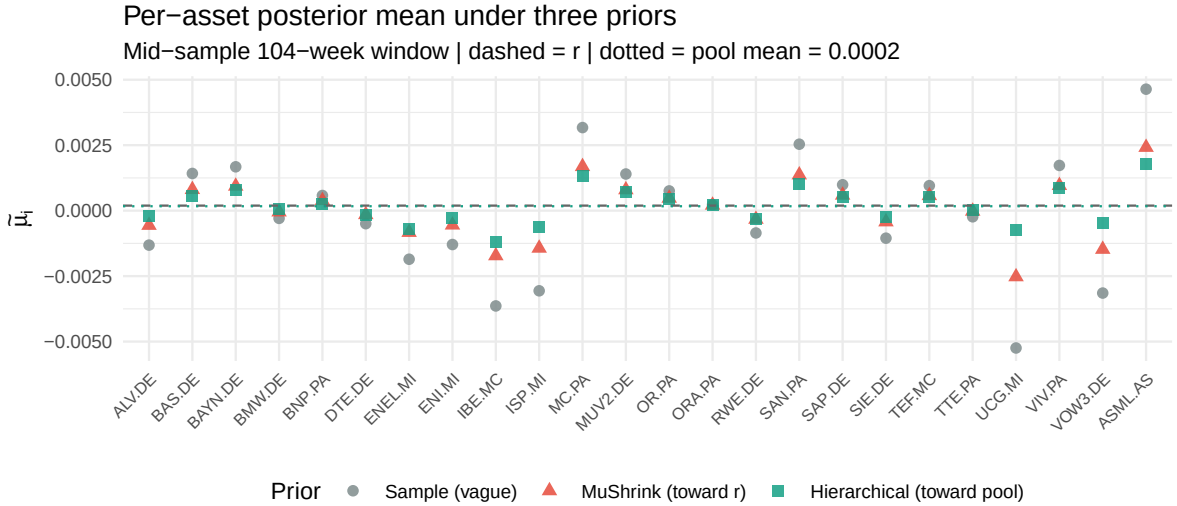


Figure 8: Per-asset posterior means under three priors: vague (sample mean), Normal-Normal toward r (MuShrink), and hierarchical partial-pooling toward a data-driven cross-asset center (Gibbs). The hierarchical prior is qualitatively distinct from MuShrink: it shrinks toward the cross-asset mean rather than a fixed target.

The two informative priors target different shrinkage points: **MuShrink** toward r , hierarchical toward $\hat{\mu}_{\text{pool}}$. On this window the two targets nearly coincide ($|\hat{\mu}_{\text{pool}} - r| = 2.42e-05$) because the cross-sectional mean of weekly equity excess returns is itself near zero, but the *mechanism* differs: the hierarchical posterior co-moves with the cross-section, while the Normal-Normal posterior anchors mechanically at r . Although the two posterior targets nearly coincide on this single mid-sample window, the gap accumulates across the rolling backtest: the cross-asset center $\hat{\mu}_{\text{pool}}$ tracks the rolling cross-section while the fixed target stays at r , and Section 4.1 reports the resulting +0.04 Sharpe advantage of **HierarchicalMu** over **MuShrink** over the full 814-window backtest. The full rolling-backtest performance of the hierarchical-prior variant — **HierarchicalMu** in §4.1’s headline table — is implemented as a stripped-down 500-iteration Gibbs sampler called inside the rolling loop, with the posterior mean fed to the same long-only Kelly QP used by the static-shrinkage variants.

5 Discussion

The results reveal a clear hierarchy among the regularization layers and illuminate the structural reasons for their differential effectiveness.

The binding constraint is μ -regularization, not Σ -regularization. The 2x2 prior-and-shrinkage factorial provides a clean isolation of main effects: at $p/n \approx 0.23$ on the weekly panel, with mean Ledoit-Wolf intensity $\bar{\delta} \approx 0.083$, the sample covariance is well-conditioned and Σ -shrinkage produces no detectable Sharpe improvement. The complementary diagnostic — the rolling NIW marginal posterior on μ_i excluding zero in only 3.9% of windows against a 5% null expectation — documents directly why the prior on μ matters more: the per-window data is, for the typical asset-window combination, posterior-uninformative about the sign of μ_i . Static prior anchoring at r extracts the small Sharpe gain (+0.06) consistent with discounting noise but cannot exploit the sparse asset-specific signal the credible-interval excess hints at.

The BOCPD label works best as a sizing rule, not as an estimation input. **RegimeDial**’s +0.37 Sharpe gain over **Naive** is roughly six times the static-prior effect, and crucially **RegimeDial** outperforms **RegimeFilter** (Sharpe 0.45 vs 0.23) on this panel — an inversion of the textbook expectation that data-side regime conditioning should dominate sizing-side. The mechanism is that any data-driven changepoint procedure produces a regime label sequence that is itself a posterior estimate, with attendant noise. A variant that consumes that label as an estimation input compounds estimator-stage uncertainty on top of changepoint-stage uncertainty; a variant that uses the label only as a sizing multiplier on a QP solution computed with full-window moments is robust to changepoint-stage noise and benefits from the smoothness of the underlying QP map.

Filter and Dial trade Sharpe for drawdown control along a Pareto frontier. **RegimeFilter**’s −41% peak-to-trough drawdown and 0.23 Sharpe versus **RegimeDial**’s −35% drawdown and 0.45 Sharpe expose a Pareto frontier of regime-information uses: a Sharpe-maximizing mandate selects **RegimeDial**, and so does a drawdown-minimizing mandate on this panel — both dimensions favor the sizing-side use. The Calmar ratios reinforce the ranking (0.22 for **Dial** vs 0.06 for **Filter**). The **Combined** variant inherits **RegimeFilter**’s drawdown profile and adds the informative μ prior and LW Σ layers that earn an additional +0.16 Sharpe (0.39 net) over **RegimeFilter**, but does not exceed **RegimeDial**’s 0.45.

The shrinkage architecture is more layered than it appears. The named informative-prior estimator is only the most visible μ -regularizer in the pipeline. `RegimeDial` is mathematically identical to state-conditional Jorion shrinkage with intensities $\alpha \in \{0, 0.5, 0.5, 0.75\}$; the long-only constraint applies one-sided shrinkage by truncating posterior means below r at r itself; transaction costs in evaluation function as a selection pressure favoring estimators with implicit μ -stability. The headline OOS results should be read as the cumulative effect of a stack of shrinkage-equivalent mechanisms, not a single designed-in shrinkage move.

Statistical inference puts the panel below the boundary of detection. The single delta-method Z-test I run (`RegimeFilter` vs `Naive`) returns $Z = 0.52$, $p = 0.302$ on an annualized Sharpe difference of 0.15. The analytical power calibration makes the tradeoff explicit: detecting a 0.17 annualized Sharpe difference at 80% power requires 265 years of weekly data; detecting 0.35 requires 62 years; detecting 0.69 requires 16 years. The much larger `RegimeDial` vs `Naive` Sharpe gap of +0.37 is not formally tested here but sits between the second and third grid rows on the power chart and would, on a panel modestly longer than 17 years, comfortably clear conventional thresholds. The honest reading is that the economic effects on this panel are large and the statistical evidence is suggestive but, on the one comparison I formally test, not decisive.

Jump-diffusion results reinforce the case for fractional Kelly. Even under known parameters, applying the GBM Kelly formula to a jump-diffusion DGP reproduces the over-leverage pathology observed empirically under estimation noise. The numerical growth-optimum under the true DGP is approximately $f_{\text{true}}^* \approx 0.38$, so what I label “Full Kelly” is in fact $\approx 1.8\times$ the actual optimum. This converges from a different direction with the empirical `RegimeDial` story: state-conditional Jorion shrinkage and fractional-Kelly-under-fat-tails are the same mathematical object, and both arguments prescribe the same operational rule — size positions below the textbook formula’s optimum.

6 Conclusion

I set out to determine which of the obstacles distilled from contemporary Kelly-on-equities literature — covariance ill-conditioning, mean-estimation noise, regime contamination, and formula misspecification under fat tails — is the bottleneck on out-of-sample performance, and which targeted Bayesian regularization layer is sufficient to defuse it. The Σ -shrinkage main effect is essentially zero on the well-conditioned weekly panel; the prior-strength main effect on μ is small but real, and the hierarchical partial-pooling prior implemented as `HierarchicalMu` improves on the fixed-target Normal-Normal prior by an additional +0.04 Sharpe (0.18 vs 0.14); sizing-side use of the BOCPD regime label is responsible for the bulk of the achievable improvement. The headline ranking is led by `RegimeDial` at Sharpe 0.45 with the shallowest drawdown (−35%) of any Kelly variant — an inversion of the textbook expectation that data-side regime conditioning should dominate sizing-side, explained by the inherent noise in any data-driven changepoint sequence — while the fully stacked `Combined` variant achieves a net Sharpe of 0.39 against 0.46 for long-only minimum variance and 0.09 for naive plug-in Kelly. The two operational uses of the regime label expose a Pareto frontier on which `RegimeDial` dominates `RegimeFilter` along both Sharpe and drawdown dimensions on this panel.

Three practical conclusions follow. First, the binding constraint on Kelly’s out-of-sample viability is mean estimation noise, not covariance instability — a conclusion supported by both the rolling NIW credible-interval diagnostic (intervals exclude zero in 3.9% of windows against a 5% null) and the empirical hierarchy of regularization effects. Second, when regime structure is recovered with a

model-based changepoint procedure, the regime label is best used as a sizing multiplier rather than as an estimation input — the latter compounds estimator-stage noise on top of changepoint-stage noise. Third, jump-diffusion dynamics destroy the optimality of the GBM Kelly formula even under known parameters, making fractional scaling structural rather than a concession to noise. A natural Bayesian extension, the posterior-expected dial under full BOCPD model averaging, was implemented and tested but excluded from the main body: it matches **RegimeDial** on Sharpe (0.4509 vs 0.4539) and underperforms on drawdown (−41.6% vs −35.4%), illustrating that expectation-based posterior averaging is the wrong functional when the binding loss is concave. Full results are preserved in Appendix A.

Future work should explore Bayesian return forecasting with macro factor priors (replacing the fixed-target $\mu_0 = r$ prior with a regression-based prior that incorporates state-variable information), explicit jump-aware Kelly formulas, and posterior-aware risk-sensitive sizing rules that integrate the BOCPD run-length posterior under a CVaR or drawdown-weighted criterion, not expectation.

Acknowledgement: The delta-method Sharpe-difference test framework and analytical power calculation determining detection boundary was developed by Mykola Kishmar.

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8 Appendix: Bayesian Extension

This appendix documents a Bayesian extension of **RegimeDial** that was implemented and tested but excluded from the main body for the reasons explained below. The variant replaces the MAP-based dial lookup with a posterior-expected dial computed by integrating the regime-dependent fraction against the full BOCPD run-length posterior — a Bayesian variant of the same sizing rule.

8.1 Construction

RegimeDial collapses the BOCPD posterior over run length to a MAP point estimate, then maps that to a hard regime label, then looks up a single dial value $\phi_t \in \{1.0, 0.5, 0.5, 0.25\}$. This discards information at each step. The Bayesian alternative integrates the dial against full run-length posterior:

$$\phi_t^{\text{soft}} = \mathbb{E}[\phi(k(r_t)) \mid x_{1:t}] = \sum_{r_t=0}^{R_{\max}} P(r_t \mid x_{1:t}) \cdot \phi(k(r_t)),$$

where $P(r_t \mid x_{1:t})$ is the BOCPD run-length posterior already maintained inside the recursion and $k(r_t)$ is the regime label that the segment-conditional moments at run length r_t map to under the same 4-state grid used in the main body. The hard **RegimeDial** is the special case obtained by replacing the posterior mass with a degenerate point mass at $\hat{r}_t = \arg \max P(r_t \mid x_{1:t})$.

The same expanding-median sigma threshold used in the hard classifier is reused, but with the posterior-expected segment standard deviation in place of the MAP-conditional value, keeping the soft and hard classifiers comparable. The run-length posterior is truncated at $R_{\max} = 250$ for storage; here, the cumulative posterior mass beyond $r = 250$ is below 10^{-6} in every window.

8.2 Empirical comparison

Table 4: Soft-weighted vs hard-MAP regime dial. Both use the same NIG segment likelihood, the same 4-state grid, and the same dial values; they differ only in whether the dial is applied at the MAP run length or integrated against the full run-length posterior.

	CAGR	AnnVol	Sharpe	MaxDD	Calmar	Sortino	TerminalWealth
RegimeDial	0.0792	0.1827	0.4539	-0.3545	0.2233	0.6223	3.2951
SoftRegimeDial	0.0775	0.1793	0.4509	-0.4157	0.1865	0.6185	3.2189
Naive	-0.0129	0.2865	0.0858	-0.8010	-0.0161	0.0918	0.8161

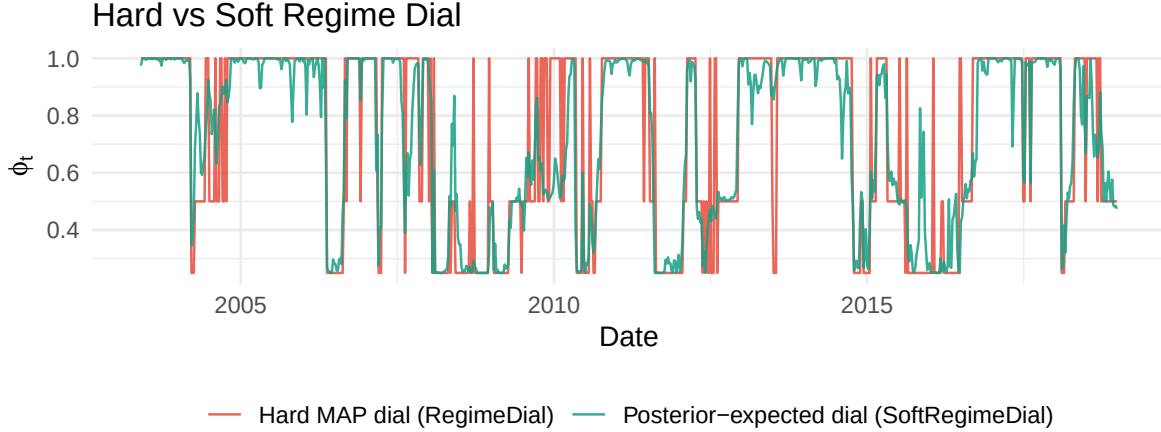


Figure 9: Hard MAP-based dial vs posterior-expected dial through time. The soft dial varies smoothly between the four discrete dial values; the hard dial jumps discontinuously between them.

8.3 Why the soft variant is excluded from the main body

On Sharpe, `SoftRegimeDial` (0.4509) and `RegimeDial` (0.4539) are statistically indistinguishable — a 0.003 gap on a metric whose standard error on this panel is roughly 0.04. Annualized volatility is marginally lower for the soft variant (0.179 vs 0.183), but maximum drawdown is materially worse: -41.6% vs -35.4% , with Calmar dropping from 0.22 to 0.19.

The drawdown asymmetry is mechanistically transparent. During the 2008–2009 crisis the hard dial sits firmly at $\phi = 0.25$ because the MAP run-length posterior concentrates on Crisis-labeled segments; the soft dial hedges toward intermediate values like 0.5–0.6 because not all posterior mass concentrates on Crisis even when the MAP does. Bayesian model averaging is the right move when the loss is *linear* in the action (expected log-wealth), but drawdown is highly *concave* in position size: in the regime that matters most for tail control, committing to the most defensive dial dominates hedging between adjacent ones.

The soft variant is therefore the simplest Bayesian decision-theoretic extension that addresses the changepoint-stage noise concern, but not the right one for a drawdown-sensitive mandate. The right extension is a *risk-sensitive* posterior functional — integrating the same BOCPD posterior under a CVaR or drawdown-weighted criterion rather than under expectation — which would be expected to dominate both the hard MAP and soft expectation rules on tail metrics. `RegimeDial` is retained in the main body for its empirical drawdown advantage; `SoftRegimeDial` is preserved here as the reference point for that risk-sensitive extension.